

Reviewer-style self-assessment of the kodama-cpp JMLR manuscript

Internal review memo and addressed revisions

Prepared with the current manuscript.

Reviewer Summary

The manuscript now reads as a software-methods submission rather than a checklist. It formalizes KODAMA as cross-validated accuracy maximization, separates raw CV accuracy from proposal acceptance scores and external diagnostics, compares the portable C++ library with the public R-package literature, and distinguishes methodological continuity from implementation novelty. The KODAMA.matrix procedure is specified with label-vector search mechanics, proposal, temperature, M/Tcycle, landmark, splitting, constraint, graph-correction rules, and explicit pseudocode at both the single-run and matrix-ensemble levels. The revision adds literature context from semi-supervised learning, weak supervision, cluster validation, circular-analysis warnings, visualization, and open-source ML reproducibility. Implementation claims are mapped to source locations, tests, and current benchmark evidence. The revision also narrows the graph-input KODAMA claim: graph input is documented and wrapper-validated as an optional API, while graph-only PLS-LDA is not presented as equivalent to data-input PLS-LDA without matched benchmarks.

Major Comments

1. Legacy KNN comparison: a matched MetRef benchmark reports the KNN-capable KODAMA 2.4.1 release and current single-core CPU, four-core CPU, and CUDA paths. It is explicitly labeled as a legacy KNN-to-KNN comparison, not as the current CRAN baseline. A separate comparison with current CRAN KODAMA 3.3 remains necessary and must disclose that its automatic PLS-DA path is not classifier-parity with latent-space PLS-LDA.
2. Continuous integration and coverage: the manually executed CPU, CUDA, Metal, R, and Python checks are substantial, but the repository does not yet contain a cross-platform CI workflow or a measured source-coverage report. Both remain release work under the JMLR MLOSS review criteria.
3. User documentation: installation and wrapper references are present, but the standalone C++ and Python interfaces still need one compact end-to-end tutorial and a browsable generated API site.
4. Public API and release identity: version 0.1.0, semantic-versioning policy, wrapper versions, and a compile-link API snapshot test are complete. The final tag, commit, source archive, and DOI remain external release actions and are not fabricated.
5. Licensing and provenance: pinned origins, SPDX identifiers, retained license texts, checksums, and the Metal Apache-2.0 exception are frozen in release documents. The residual coauthor-contribution confirmation is stated explicitly.
6. MLOSS format: the official-style main paper now occupies four description pages plus references; the detailed manuscript is retained as a technical supplement.
7. Visualization evidence: a fixed CUDA M=100/Tcycle=100 panel now reports both classifiers, both embeddings, runtime, ARI, active classes, silhouette, and local purity on four named nonspatial datasets. Its positive and negative outcomes are retained. Larger-scale and matched backend/historical-wrapper matrices remain required before broad empirical claims.

Revisions Applied

- Removed application-specific discussion and focused the paper on the KODAMA principle and implementation.
- Removed unsupported experimental paths and described only KNN and PLS-LDA as KODAMA classifiers.
- Added theory framing for cross-validated label predictability.

- Expanded the description of how high cross-validated accuracy label vectors are found: CV predictions guide group proposals, class-transition proposals aggregate unstable label movement, one CV pass evaluates each proposal, and a cooling acceptance rule maintains exploration.
- Expanded the mathematical description into a reproducible KODAMA.matrix procedure, including M runs, Tcycle, landmark selection, splitting, constrained groups, proposal rules, temperature acceptance, and KODAMA graph correction.
- Added KODAMA pseudocode that makes the initial CV evaluation, one-CV-per-cycle proposal loop, acceptance rule, projection step, and final graph reweighting explicit.
- Added a focused single-run pseudocode block showing one independent M run as the stochastic unit repeated by the ensemble.
- Completed a pinned licensing and provenance audit with per-file SPDX identifiers, retained third-party notices and license checksums, explicit KODAMA/fastPLS relicensing scope, and the MIT/Apache-2.0 Metal exception.
- Added a public-literature comparison against the original KODAMA paper, the Bioinformatics R-package paper, and the documented R interface.
- Added a related-work section positioning KODAMA relative to semi-supervised graph learning, pseudo-labeling, weak supervision, clustering stability, prediction strength, UMAP, t-SNE, and open-source ML reproducibility.
- Added a compact architecture figure showing R/Python wrappers, the C++17 core, CPU, CUDA, and Metal backends, graph/embedding/clustering utilities, and typed outputs.
- Reframed the implementation contribution around backend-native state ownership and added separate CUDA and Apple Metal subsections with explicit host/device boundaries, reusable workspaces, strict backend identity, and a shared parity contract.
- Added implementation details on float32 storage, label-aware SIMPLS, package-owned CPU/CUDA/Metal neighbor search, independent M cycles, typed outputs, and release validation.
- Replaced the installed FAISS/cuVS CUDA search path with package-owned exact/IVF KNN and k-means, and added clean-build, CTest, soname-audit, MNIST70k, and MetRef evidence.
- Added an implementation-evidence matrix linking each claim to source files, tests, and measured benchmark rows.
- Added evaluation guardrails that explicitly address circularity, parameter dependence, visualization bias, runtime attribution, and wrapper reproducibility.
- Added an ablation matrix specifying the required classifier, graph-correction, M/Tcycle, landmark/splitting, backend, and wrapper-parity comparisons.
- Added exact benchmark protocol, dataset coverage, and installation commands for the C++ core plus R and Python wrappers.
- Added an explicit distinction between raw CV accuracy, proposal acceptance score, and external diagnostics.
- Added a frozen 0.1.0 public API contract, version header, compile-link API test, changelog, contribution guide, prior-version delta, release checklist, and clean-tag archive script.
- Separated the submission into an official four-page MLOSS paper plus references and a detailed technical supplement.
- Added a track-specific cover-letter draft and a reviewer-comment response matrix with all author-only and external release fields left explicit.
- Added resumable release-validation, nonspatial M/Tcycle sensitivity, historical-R, backend, and six-panel UMAP/openTSNE benchmark drivers.
- Inserted completed current-validation results from the CUDA workstation: CV kernel speedups, CoreKNN/CorePLSLDA optimization medians, and the MetRef KODAMA.matrix validation benchmark.
- Completed the named MetRef/USPS M/Tcycle sensitivity experiment on CUDA and inserted the numerical tables and curve figure.
- Added an agreement-graph convergence analysis showing why M controls Monte Carlo precision rather than single-run optimization quality; reported empirical RMSE/correlation and the $1/(2 \sqrt{M})$ worst-case standard-error bound.
- Updated the end-to-end MetRef evidence to M=100/Tcycle=100 and reported the KNN collapse alongside the much stronger PLS-LDA external-label and visualization diagnostics.

- Replaced the obsolete short-run visualization preview with a reproducible four-dataset CUDA panel at $M=100/T_{\text{cycle}}=100$, archived its complete CSV and six-panel plots, and added a balanced silhouette/runtime/ARI figure that retains negative results.
- Completed the matched MetRef legacy KNN comparison: current single-core, four-core, and CUDA wall times are reported against KODAMA R 2.4.1, with explicit disclosure that it is not the current CRAN release and that proposal trajectories differ.
- Removed draft-status boxes and footer language from the manuscript.

Remaining Work

- Extend the completed four-dataset visualization panel to larger nonspatial matrices and run the remaining matched backend and release-ablation matrices on the release hardware.
- Add Linux and macOS continuous-integration jobs for the standalone core and wrappers, retain accelerator validation as a separately documented hardware job, and publish a measured line/branch coverage report.
- Publish a compact C++/R/Python end-to-end tutorial and generated API documentation at a stable project URL.
- Create and push the clean v0.1.0 tag, archive it, record the SHA-256 digest, and deposit it for an archival DOI.
- Obtain coauthor consent; complete funding, competing-interest, action-editor, reviewer, and current user-community fields in the cover letter; and confirm that no adapted KODAMA or fastPLS code was supplied outside the inspected Git history, or retain the relevant permission.

Reviewer Recommendation

Recommendation: promising JMLR MLOSS submission after the listed release-engineering actions. The four-page paper is focused, the KODAMA search is reproducible from the text, the heterogeneous architecture is a non-trivial software contribution, and limitations and negative results are stated with unusual clarity. Additional theory is not the priority.

The principal desk-review risks are the absence of committed cross-platform CI and measured coverage, the lack of a stable hosted tutorial/API site, and incomplete tag, DOI, community-metric, and author declarations. Broader benchmarks strengthen impact, but these software-quality items are at least as important under the published MLOSS criteria.